

Computational Finance (WI4151) — Oral Exam: Worked Answers, Part 2

20 more concept-level questions and answers, English. Companion to
`oral-exam-questions-solved.pdf`.

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How to use this. Part 1 covered the 11 questions reported from the 23 June oral (the three methods, Bermudan by MC/COS, correlated paths, FD schemes, the COS breakthrough, CDF inversion, affine cf, the pricing PDE, the COS steps). This Part 2 adds 20 *more* questions of the same conceptual flavour, spanning the whole course. Same rule as the real exam: answer the *idea*, then go one layer deeper if pushed.

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1 Risk-neutral pricing and the Black–Scholes world

Q1. Why do we price under the risk-neutral measure $\mathbb{Q}$ and not the physical measure $\mathbb{P}$? What is Girsanov’s role?

A. Because a price is the *cost of a hedge*, not a forecast. By no-arbitrage, a contingent claim that can be replicated must cost the same as its replicating portfolio, and that cost equals the discounted expectation under the *equivalent martingale measure* $\mathbb{Q}$ — the measure under which discounted tradable prices are martingales. Under the physical measure $\mathbb{P}$ the stock drifts at μ , which encodes risk preferences; delta-hedging removes the risk, so preferences (and μ) cannot enter the price. *Girsanov* is the tool that changes measure: it shifts the Brownian drift by the *market price of risk* $\theta = (\mu - r)/\sigma$, turning $dW^{\mathbb{P}}$ into $dW^{\mathbb{Q}} = dW^{\mathbb{P}} + \theta dt$, so the stock drift becomes r while the volatility σ (the quadratic variation) is unchanged. Equivalence ($\mathbb{P} \sim \mathbb{Q}$, same null sets) is what keeps the model arbitrage-free.

Q2. What exactly is the characteristic function, and why is it the central object of this course?

A. The characteristic function of a random variable X is $\varphi_X(u) = \mathbb{E}[e^{iuX}] = \int_{\mathbb{R}} e^{iux} f(x) dx$ — the Fourier transform of the density. It *always exists* ($|e^{iuX}| = 1$), it *uniquely determines* the distribution, and its log generates the cumulants ($c_n = \frac{1}{i^n} \frac{d^n}{du^n} \ln \varphi_X(u) \big|_{u=0}$). It is central because for almost every model used in pricing (Black–Scholes, Heston, Variance-Gamma, Merton/Kou jumps, affine jump-diffusions) the *density is intractable but φ is known in closed form*. COS turns

that knowledge directly into a price (the cosine coefficients of f are $\text{Re}\{\varphi\}$ on a grid), and the cumulants of φ fix the COS truncation range.

Q3. Why does COS use a *cosine* series and not a full complex Fourier series?

A. Three reasons: (i) extending f to an *even* function on $[a, b]$ kills all the sine terms, so the coefficients are *real* — real arithmetic, half the storage; (ii) the cosine coefficient is exactly the *real part* of the characteristic function at the grid frequency $u_k = k\pi/(b-a)$, so it is read off φ in one line with no inversion integral; (iii) the payoff coefficients V_k then reduce to *elementary closed forms* (χ_k, ψ_k). A full complex Fourier series carries twice the coefficients and complex bookkeeping for no benefit.

Q4. Why is the COS truncation range chosen as $[a, b] = c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$? Interpret each term.

A. The range must contain (almost) all the probability mass of the log-return. c_1 (the mean) *centres* the interval; c_2 (the variance) sets the natural width scale $\sqrt{c_2}$, so $L\sqrt{c_2}$ is an “ L -standard-deviation” box; $\sqrt{c_4}$ (from the fourth cumulant, i.e. kurtosis) *fattens* the box for heavy-tailed models and vanishes for Gaussian returns; $L \approx 8\text{--}12$ is the safety multiplier. The trade-off: too narrow $\Rightarrow$ you truncate real mass and the price is biased (a fixed error floor); too wide $\Rightarrow$ you need more terms N for the same accuracy (cost grows with $b-a$).

Q5. What governs the convergence rate of COS — when is it exponential and when only algebraic?

A. There are three error sources: *range truncation* (fixed once $[a, b]$ is chosen — it sets the accuracy floor), *series truncation* at N , and any error in φ . The rate in N is governed by the decay of the cosine coefficients A_k , which is governed by the *smoothness* of the density: if f is analytic (or C^∞ with integrable derivatives) the A_k decay *exponentially* $\Rightarrow$ exponential convergence (machine precision in dozens–hundreds of terms); if f has only finite smoothness (β integrable derivatives) then $A_k = O(k^{-(\beta+1)}) \Rightarrow$ only *algebraic* convergence. So a *lack of smoothness* is what kills the exponential rate — e.g. Variance-Gamma with small ν (a cusp in the density) or a very short maturity (the density approaches a Dirac).

Q6. Compare COS with the Carr–Madan FFT method. When does COS win?

A. Both need only the characteristic function. Carr–Madan damps the call ($c_T(k) = e^{\alpha k} C_T(k)$ in log-strike), Fourier-transforms it analytically, and *inverts numerically by FFT*. Its weaknesses, all of which COS removes: it performs a numerical *integration* (so it is quadrature-limited, only algebraic accuracy, and needs thousands of FFT nodes), and it needs a free *damping parameter* α to make the call integrable. COS is one $O(N)$ cosine sum — *no integration, no FFT, no α* — with *exponential* convergence, reaching machine precision in a few dozen to a few hundred terms. COS wins for smooth, low-dimensional European problems with a known φ ; you only have to choose $[a, b]$ carefully.

2 Stochastic volatility and jumps

Q7. What is the Feller condition in the Heston model and why does it matter?

A. Heston’s variance follows the CIR process $dv = \kappa(\bar{v} - v)dt + \eta\sqrt{v}dW$. The Feller condition $2\kappa\bar{v} \geq \eta^2$ guarantees that the variance stays strictly positive (the origin is *unattainable*). It matters in two places: (i) *simulation* — if it is violated, v visits zero, the $\sqrt{v}$ diffusion degenerates and an Euler scheme produces negative variance; (ii) *the density / cf* — when v piles up at 0 the conditional density is badly behaved and the characteristic-function evaluation (hence COS) loses accuracy. In words: it is the condition under which the mean-reversion “push” away from zero ($2\kappa\bar{v}$) dominates the noise (η^2).

Q8. Why is Heston *not* affine in (S, v) but *is* affine in $(\ln S, v)$?

A. “Affine” means the drift, the instantaneous covariance and the jump intensity are all linear-plus-constant in the state. In (S, v) the stock diffusion is $\sqrt{v} S dW$, so the variance of S is vS^2 — it involves the *product* $v \cdot S^2$, which is not affine. Switch to the log-price $x = \ln S$: $dx = (r - \frac{1}{2}v) dt + \sqrt{v} dW_1$. Now the drift $r - \frac{1}{2}v$ is affine in v , the variance of x is v (affine in v), and the variance dynamics are already affine. So the pair (x, v) is an affine process, which is exactly why its characteristic function is exponential-affine $\varphi = \exp(\alpha + \beta \cdot (x, v))$ obtained from Riccati ODEs.

Q9. How do you simulate the Heston model, and what goes wrong with a naive Euler scheme?

A. The problem is the variance leg. A plain Euler step $v_{n+1} = v_n + \kappa(\bar{v} - v_n)\Delta t + \eta\sqrt{v_n}\Delta W$ can overshoot below zero (especially when Feller is violated), and then $\sqrt{v_{n+1}}$ is undefined. Fixes: *full truncation* (use $v^+ = \max(v, 0)$ inside the drift and diffusion), *reflection* ($|v|$), or the log-variance / Andersen-QE schemes. The *exact* method (Broadie–Kaya) samples v_T from its noncentral- χ^2 law and the *integrated variance* $\int v ds$ from its characteristic function (numerically inverting the recovered CDF), then samples S_T given those. Trade-off: exact is unbiased but slow; truncation schemes are fast but introduce a discretisation bias. The spot and variance Brownian motions are correlated by ρ (Cholesky, Part 1 Q5).

Q10. What is the difference between strong and weak convergence of an SDE scheme, and which one matters for option pricing?

A. *Strong* convergence controls the pathwise error: $\mathbb{E}|X_T^{\Delta t} - X_T| \leq C \Delta t^p$ (order p). *Weak* convergence controls the error in distribution / in expectations: $|\mathbb{E}g(X_T^{\Delta t}) - \mathbb{E}g(X_T)| \leq C \Delta t^q$ (order q). A European price is an *expectation* $\mathbb{E}[\text{payoff}]$, so the *weak* order is what governs the bias (Euler already has weak order 1). Strong order matters when paths themselves matter — path-dependent payoffs, or multilevel Monte Carlo. Euler–Maruyama is (strong $\frac{1}{2}$, weak 1); Milstein is (strong 1, weak 1).

Q11. What is a jump-diffusion / compound Poisson process and what is its characteristic function?

A. A compound Poisson process is $Q_t = \sum_{i=1}^{N_t} J_i$, where N_t is a Poisson process with intensity λ (jumps arrive at rate λ) and the J_i are i.i.d. jump sizes. Conditioning on N_t and using the Poisson law gives its characteristic function in closed form,

$$\varphi_{Q_t}(u) = \exp(t \lambda (\theta_J(u) - 1)), \quad \theta_J(u) = \mathbb{E}[e^{iuJ}],$$

where $\lambda(\theta_J(u) - 1)$ is the *Lévy–Khintchine jump exponent*. A jump-diffusion log-price adds a Brownian part, so $\varphi(u, t) = \exp(t[iu\mu - \frac{1}{2}\sigma^2 u^2 + \lambda(\theta_J(u) - 1)])$. Merton takes Gaussian jumps (θ_J a Gaussian cf); Kou takes double-exponential jumps. Because φ closes, COS prices these directly. (Risk-neutrality requires a drift compensator $\mu = r - \lambda\mathbb{E}[e^J - 1]$.)

Q12. What is the PIDE for a jump model and why is it “non-local”? Why prefer Fourier methods?

A. For a jump-diffusion the value solves a *partial integro-differential equation*: besides the usual diffusion terms, the generator has an integral term $\lambda \int [V(x+z, t) - V(x, t) - (e^z - 1)\partial_x V] \nu(dz)$. It is *non-local* because a jump connects the point x to far-away points $x+z$ — the equation at x couples to the solution everywhere, not just to neighbouring grid points. Finite differences must then handle a dense/non-local operator at every step, which is expensive and awkward. Fourier/COS methods avoid this entirely: the jump enters *analytically* through the characteristic function, so the same $O(N)$ machinery applies with no extra cost. This is a major reason Fourier methods dominate for Lévy/jump models.

3 PDE / finite-difference and lattice methods

Q13. Write the Black–Scholes PDE and explain what each term means, with the boundary and terminal conditions.

A. The PDE is

$$\partial_t V + rS \partial_S V + \frac{1}{2} \sigma^2 S^2 \partial_{SS} V - rV = 0.$$

Reading the terms at a fixed spot: $\partial_t V$ is time decay (theta); $rS \partial_S V$ is the risk-neutral drift acting on the delta; $\frac{1}{2} \sigma^2 S^2 \partial_{SS} V$ is the diffusion / convexity term (gamma) — the only place volatility enters; $-rV$ is discounting. The *terminal* condition is the payoff, $V(S, T) = g(S)$ (it is solved *backward* in time). *Boundary* conditions at the edges of the S -domain, e.g. for a call $V(0, t) = 0$ and $V(S, t) \rightarrow S - Ke^{-r(T-t)}$ as $S \rightarrow \infty$; for a knock-out barrier, a homogeneous (absorbing) condition at the barrier. It is exactly the Feynman–Kac PDE of the risk-neutral expectation.

Q14. Which finite-difference schemes are there, and how do you choose between them?

A. On the heat-equation form, with $\nu = \Delta t / \Delta x^2$: *FTCS* (explicit) — cheap, one-step update, but only conditionally stable ($\nu \leq \frac{1}{2}$ by von Neumann, so the time step must scale like Δx^2); *BTCS* (implicit) — solve a tridiagonal system each step, *unconditionally* stable; *Crank–Nicolson* (their average) — second-order accurate in time ($O(\Delta t^2 + \Delta x^2)$) and unconditionally stable, the usual default, with mild oscillations possible near non-smooth payoffs. They are the $\theta = 0, 1, \frac{1}{2}$ members of the θ -scheme. Choose CN for accuracy; FTCS only when simplicity matters and the CFL step is affordable; for American options replace the linear solve by projected SOR (PSOR).

Q15. Explain von Neumann stability and the CFL condition for the explicit scheme.

A. Insert a Fourier mode $u_j^i = \xi^i e^{i\beta j \Delta x}$ into the scheme and solve for the *amplification factor* $\xi(\beta)$; the scheme is stable iff $|\xi(\beta)| \leq 1$ for all modes β . For FTCS one gets $\xi = 1 - 4\nu \sin^2(\beta \Delta x / 2)$, and $|\xi| \leq 1$ forces the *CFL condition* $\nu = \Delta t / \Delta x^2 \leq \frac{1}{2}$, i.e. the time step must shrink like the square of the space step — expensive when refining the grid. For BTCS $\xi = 1 / (1 + 4\nu \sin^2(\beta \Delta x / 2))$, which is ≤ 1 for every ν , so it is unconditionally stable. (Lax equivalence: consistency + stability $\Rightarrow$ convergence.)

Q16. How is the binomial tree related to finite differences, and how are the CRR parameters chosen?

A. The CRR up/down factors u, d and risk-neutral probability p are fixed by *matching the first two moments* (mean and variance) of the one-step log-return to the risk-neutral GBM values; pricing is then backward induction $V = e^{-r\Delta t}(pV_u + (1-p)V_d)$. This is precisely an *explicit finite-difference scheme on a log-price grid*: with the coupling $\Delta x^2 = \sigma^2 \Delta t$ the central node of the three-point explicit stencil drops out and you are left with the two-point binomial recursion. So a binomial tree is an explicit FD scheme specialised to a single time-zero value; convergence is $O(1/M)$ and non-monotone (the error oscillates as the strike straddles nodes).

Q17. Why do we work in $x = \ln S$ in both finite differences and COS?

A. Because it makes the coefficients *constant*. In S the BS PDE has S -dependent coefficients ($rS \partial_S, \frac{1}{2} \sigma^2 S^2 \partial_{SS}$); on a uniform S -grid the diffusion weight grows like j^2 , which hurts conditioning and stability at large S . In $x = \ln S$ the PDE becomes a constant-coefficient advection–diffusion equation $\partial_t V + (r - \frac{1}{2} \sigma^2) \partial_x V + \frac{1}{2} \sigma^2 \partial_{xx} V - rV = 0$, which is uniform, well-conditioned and reduces cleanly to the heat equation. COS likewise works in the log-state because the characteristic functions and the cumulant range are naturally expressed for the log-return. (It is the same domain lesson as the COS truncation $[a, b]$.)

4 Foundations, Greeks and variance reduction

Q18. State put–call parity, explain why it is model-free, and how it is used numerically.

A. Put–call parity is $C + Ke^{-r(T-t)} = P + S_t$. It follows from a *static* replication: a portfolio of a call plus cash $Ke^{-r(T-t)}$ and a portfolio of a put plus the share both pay $\max(S_T, K)$ at maturity, so by no-arbitrage they cost the same today. It uses *only* no-arbitrage and a constant rate — no assumption about the dynamics of S — hence it is *model-free*. Numerically it gives a free correctness check on any pricer, lets you recover one IV from the other (a put and call at the same (K, T) share one implied vol), and its barrier analogue — *in-out parity* $C_{\text{in}} + C_{\text{out}} = C_{\text{vanilla}}$ — lets you price the easier knock-out leg and subtract.

Q19. Why is an American call on a non-dividend stock never exercised early, while the put can be?

A. For the call, the no-arbitrage bound $C \geq S - Ke^{-r(T-t)} > S - K$ (for $r > 0$) says the *live* call is worth strictly more than its intrinsic value $S - K$, so you would never throw away that surplus by exercising early — hence American call = European call. Intuitively, exercising early forfeits the interest you could earn on the strike and the remaining “insurance” (time) value. For the *put*, parity gives $P \geq Ke^{-r(T-t)} - S$, and since $Ke^{-r(T-t)} < K$ the live put can fall *below* its intrinsic value $K - S$; then exercising early to receive K now and earn interest on it can be optimal. That is why the put has a genuine free boundary and the call does not (without dividends).

Q20. What is the reflection principle, and how does it give the law of the running maximum used to price barriers under GBM?

A. For a standard Brownian motion W , the events $\{\max_{[0,T]} W \geq a\}$ and $\{\tau_a \leq T\}$ (first hitting of level a) coincide. Reflecting the path after τ_a shows that a path which has touched a is equally likely to end above or below a , giving $\mathbb{P}(\tau_a \leq T) = 2\mathbb{P}(W_T \geq a)$. Hence the running maximum has the law of $|W_T|$, with density $p_{\max}(a) = \sqrt{2/(\pi T)} e^{-a^2/2T}$ on $a \geq 0$; reflecting the endpoint at $2a - b$ gives the *joint* density of $(\max, W_T)$. Under GBM the barrier-crossing event is controlled by the maximum of a *drifted* Brownian motion, and integrating the payoff against this joint density (the drift handled by a Girsanov reweighting) yields the closed-form barrier prices. This is the analytic backbone behind the (continuously-monitored) barrier formulas.

Greeks across the three methods (one compact answer worth memorising).

- **Monte Carlo:** *pathwise* derivative (differentiate the discounted payoff along the path), the *likelihood-ratio* method (differentiate the density/weight, good for discontinuous payoffs), or *bump-and-revalue* finite differences with *common random numbers* to control the noise.
- **Finite differences:** read the Greeks straight off the grid — $\Delta = \partial_S V$ and $\Gamma = \partial_{SS} V$ from neighbouring nodes, Θ from adjacent time levels — essentially free.
- **COS:** differentiate the closed-form sum *analytically*. Since S_0 enters only through the cf factor e^{iux_0} , Δ and Vega are termwise derivatives of the same $O(N)$ sum — fast and high-accuracy, no re-integration.

Closing one-liner (ties Parts 1 and 2 together). Everything is one identity — $V = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{payoff}]$ — attacked three ways: *sample* it (Monte Carlo), *solve the PDE* it satisfies (finite differences), or *evaluate the integral via the characteristic function* (COS); the same cf / affine structure, Cholesky correlation, reflection principle and backward recursion then recur across Heston, jumps, American/Bermudan and barrier options.